Maths Revision Notes

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Contents

1 Pure Year I **1**

1.1 Algebraic Expressions . . . 1
1.2 Quadratics . . . 2
1.2.1 The Discriminant . . . 2
1.2.2 Factorising . . . 2
1.2.3 Quadratic Inequalities . . . 3
1.2.4 Set-Builder Notation . . . 3
1.2.5 Completing the Square . . . 4
1.2.6 Quadratic Turning Points . . . 4
1.3 Equations and Inequalities . . . 6
1.4 Graphs and Transformations . . . 7
1.5 Straight Line Graphs . . . 8
1.6 Circles . . . 9
1.7 Algebraic Methods I . . . 10
1.8 The Binomial Expansions . . . 11
1.9 Trigonometric Ratios . . . 12
1.10 Trigonometric Equations and Identities . . . 13
1.11 Vectors I . . . 14
1.12 Differentiation I . . . 15
1.13 Integration I . . . 16
1.14 Exponentials and Logarithms . . . 17

2 Pure Year II **19**

2.1 Algebraic Methods II . . . 19
2.2 Functions and Graphs . . . 20
2.3 Sequences and Series . . . 21
2.4 Binomial Expansion . . . 22
2.5 Radians . . . 23
2.6 Trigonometric Functions . . . 24
2.7 Trigonometry and Modelling . . . 25
2.8 Parametric Equations . . . 26
2.9 Differentiation II . . . 27
2.10 Numerical Methods . . . 28
2.11 Integration II . . . 29

2.12 Vectors II 30

3 Mechanics **31**
3.1 Mechanics Year I 31
3.1.1 Modelling in Mechanics 31
3.1.2 Constant Acceleration 32
3.1.3 Forces and Motion 33
3.1.4 Variable Acceleration 34
3.2 Mechanics Year II 35
3.2.1 Moments 35
3.2.2 Forces and Friction 36
3.2.3 Projectiles 37
3.2.4 Applications of Forces 38
3.2.5 Further Kinematics 39

4 Statistics **41**
4.1 Statistics Year I 41
4.1.1 Data Collection 41
4.1.2 Measures of Location and Spread 42
4.1.3 Representations of Data 43
4.1.4 Correlation 44
4.1.5 Probability 45
4.1.6 Statistical Distributions 46
4.1.7 Hypothesis Testing 47
4.2 Statistics Year II 48
4.2.1 Regression, Correlation and Hypothesis Testing 48
4.2.2 Conditional Probability 49
4.2.3 The Normal Distribution 50

A Appendix: Formula Sheet **51**

B Appendix: Non-Formula Sheet **53**

Chapter 1

Pure Year I

1.1 Algebraic Expressions

1.2 Quadratics

A quadratic is an expression in the form $ax^2 + bx + c$, for some real constants a, b and c. The solutions for a quadratic in the form $ax^2 + bx + c = 0$ are given by the quadratic formula:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \tag{1.1}$$

1.2.1 The Discriminant

From the quadratic formula (equation 1.1), we can see that if $b^2 - 4ac < 0$, the square root will give an imaginary number, meaning that the equation has no real roots. This part of the formula is called the discriminant, written Δ:

$$\Delta = b^2 - 4ac \tag{1.2}$$

Using the discriminant, we can tell whether a quadratic has one distinct real root, two distinct real roots, or no real roots (two complex roots).

- **No real roots:** if $\Delta < 0$, then $\sqrt{\Delta} \notin \mathbb{R} \implies x \notin \mathbb{R}$, so there are no real roots.

- **One distinct real root:** if $\Delta = 0$, then both values of x are equal, so both of the roots are the same ($x = \frac{-b}{2a}$), thus the quadratic has only one distinct real root. The Fundamental Theorem of Algebra says that for an nth degree polynomial, there exists n roots, so the single distinct root is technically two roots with the same value.

- **Two distinct real roots:** if $\Delta > 0$, then the value of $+\sqrt{\Delta}$ is not the same as the value of $-\sqrt{\Delta}$, so the two roots are both real and different (since $a \neq -a \ \forall \ a \in \mathbb{R} \setminus \{0\}$ and $\sqrt{\Delta} \neq 0$ if $\Delta > 0$), giving two distinct real roots.

1.2.2 Factorising

A quadratic $ax^2 + bx + c = 0$ can be factorised into a pair of binomials $a(x - \alpha)(x - \beta) = 0$, where α and β are the roots of the polynomial. To find α and β, find two values m and n that multiply to ac and sum to b. With these values, the expression $ax^2 + bx + c$ can be rearranged into $(ax^2 + mx) + (nx + mn)$. From these brackets, find a common value to factor out of both, and then it will simplify to a pair of binomials.

This can then be used to get the roots. Set each bracket individually to 0, and then solve for α or β.

1.2.3 Quadratic Inequalities

1.2.4 Set-Builder Notation

A set of values can be written using set-builder notation as:

$$\{x \in S : \phi(x)\} \tag{1.3}$$

Where $\phi(x)$ is a condition which can be true or false, based on the value of x, and S is a set containing x. The set represents the set of all values x in the set S that satisfy $\phi(x)$. The '$\in S$' part can be omitted if it is obvious which set is being talked about, usually $S = \mathbb{R}$.

An example of set-builder notation would be the set of all integers greater than 2:

$$\{x \in \mathbb{Z} : x > 2\} \tag{1.4}$$

This would be equivalent to the set:

$$\{3, 4, 5, \cdots\} \tag{1.5}$$

However, the condition $\phi(x)$ can be more complex; ϕ (I have removed the '(x)' for simplicity, but ϕ has the same meaning as $\phi(x)$) can be built from other conditions χ and ψ using boolean logic:

$$\chi \text{ and } \psi : \ \chi \wedge \psi \tag{1.6}$$
$$\chi \text{ or } \psi : \ \chi \vee \psi \tag{1.7}$$
$$\text{not } \psi : \ \neg \psi \tag{1.8}$$

Alternatively, they can be written using unions and intersections of multiple sets:

$$\{x : \chi \wedge \psi\} = \{x : \chi\} \cap \{x : \psi\} \tag{1.9}$$
$$\{x : \chi \vee \psi\} = \{x : \chi\} \cup \{x : \psi\} \tag{1.10}$$
$$\{x \in S : \neg \psi\} = S \setminus \{x : \psi\} \tag{1.11}$$

Set-builder notation can be used to represent solution sets. For example, the set of values that satisfy $\alpha < x < \beta$ can be written:

$$\{x \in \mathbb{R} : \alpha < x < \beta\} \tag{1.12}$$

The set of values that satisfy $x < \alpha$ or $\beta < x$ (where $\alpha < \beta$) can be written:

$$\{x \in \mathbb{R} : (x < \alpha) \vee (\beta < x)\} = \{x \in \mathbb{R} : x < \alpha\} \cup \{x \in \mathbb{R} : \beta < x\} \tag{1.13}$$

Alternatively, this set also represents all values of x in $\mathbb{R}$ that are not in the closed interval $[\alpha, \beta]$, thus could be written $x \in \mathbb{R} \setminus [\alpha, \beta]$.

For a quadratic $p(x) = 0$, the solution set would not require set-builder notation, since there are only one or two values, so it can be written $\{\alpha, \beta\}$. If there are no real roots, the set of real roots is empty, and can be written as the empty set $\emptyset = \{\}$.

1.2.5 Completing the Square

Theorem 1.2.1 (Completing the Square)**.** *A quadratic in the form ax^2+bx+c can be written in the form $a(x+m)^2+n$, where $m = \frac{b}{2a}$ and $n = -\frac{b^2}{4a} + c$:*

$$a^2 + bx + c = a\left(x + \frac{b}{2a}\right)^2 - \frac{b^2}{4a} + c \tag{1.14}$$

Proof. We will begin by factoring a out of the first two terms of the original quadratic to get:

$$ax^2 + bx + c = a\left(x^2 + \frac{b}{a}\right) + c \tag{1.15}$$

Now, the $\left(x^2 + \frac{b}{a}\right)$ part can almost be factorised into $\left(x + \frac{b}{2a}\right)^2$, however expanding that will give an extra $\left(\frac{b}{2a}\right)^2$ term. This means that it can be rewritten as:

$$\left(x^2 + \frac{b}{a}\right) = \left(x + \frac{b}{2a}\right)^2 - \frac{b^2}{4a^2} \tag{1.16}$$

This can be put back into the formula to get:

$$ax^2 + b + c = a\left(\left(x + \frac{b}{2a}\right)^2 - \frac{b^2}{4a^2}\right) + c \tag{1.17}$$

Moving the $\frac{b^2}{4a^2}$ term out of the brackets (and thus multiplying by a):

$$ax^2 + bx + c = a\left(x + \frac{b}{2a}\right)^2 - \frac{b^2}{4a} + c \tag{1.18}$$

□

1.2.6 Quadratic Turning Points

A quadratic $p(x)$ is a parabola, so has a turning point where $\frac{dp}{dx} = 0$. This can be found in one of two ways:

- **Completing the Square:** use completing the square (see section 1.2.5) to put the quadratic in the form $a(x-m)^2+n$. This is the same as using graph transformations to move the parabola ax^2 (which has a turning point $(0,0)$) right by m (subtract m from the x value), and then moving it up by n (add n to the overall value). This means that the turning point is going to be $(0,0)$, moved right by m and up by n, which is (m,n). Substituting in the values for the quadratic ax^2+bx+c, the turning point is $\left(-\frac{b}{2a}, -\frac{b^2}{4a} + c\right)$.

- **Differentiation:** since the turning point is the point where $\frac{dp}{dx} = 0$, it can be found by finding the derivative of $p(x)$, which (for $p(x) = ax^2 + bx + c$) is $2ax + b$. Then, set $2ax + b$ to 0, giving $x = \frac{b}{2a}$. This can then be substituted into $p(x)$ to give the y-coordinate ($y = p\left(\frac{b}{2a}\right)$).

1.3 Equations and Inequalities

1.4 Graphs and Transformations

1.5 Straight Line Graphs

1.6 Circles

1.7 Algebraic Methods I

1.8 The Binomial Expansions

1.9 Trigonometric Ratios

1.10 Trigonometric Equations and Identities

1.11 Vectors I

1.12 Differentiation I

1.13 Integration I

1.14 Exponentials and Logarithms

Chapter 2

Pure Year II

2.1 Algebraic Methods II

2.2 Functions and Graphs

2.3 Sequences and Series

2.4 Binomial Expansion

2.5 Radians

2.6 Trigonometric Functions

2.7 Trigonometry and Modelling

2.8 Parametric Equations

2.9 Differentiation II

2.10 Numerical Methods

2.11 Integration II

2.12 Vectors II

Chapter 3

Mechanics

3.1 Mechanics Year I

3.1.1 Modelling in Mechanics

3.1.2 Constant Acceleration

3.1.3 Forces and Motion

3.1.4 Variable Acceleration

3.2 Mechanics Year II

3.2.1 Moments

3.2.2 Forces and Friction

3.2.3 Projectiles

3.2.4 Applications of Forces

3.2.5 Further Kinematics

Chapter 4

Statistics

4.1 Statistics Year I

4.1.1 Data Collection

4.1.2 Measures of Location and Spread

4.1.3 Representations of Data

4.1.4 Correlation

4.1.5 Probability

4.1.6 Statistical Distributions

4.1.7 Hypothesis Testing

4.2 Statistics Year II

4.2.1 Regression, Correlation and Hypothesis Testing

4.2.2 Conditional Probability

4.2.3 The Normal Distribution

Appendix A

Appendix: Formula Sheet

Appendix B

Appendix: Non-Formula Sheet